THE THETA SERIES $\theta_D(z)\theta_D(pz)$ FOR THE ODD HEEGNER DISCRIMINANTS

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ABSTRACT. For the seven odd discriminants of class number one, $D \in \{3,7,11,19,43,67,163\}$, the weight-one theta series θ_D of the norm form $N_D = x^2 + xy + \frac{D+1}{4}y^2$ is the Eisenstein series $E_1(z,\chi_{-D})$, and $\Theta_{1,p}^{(D)} = \theta_D(z)\theta_D(pz)$ is a form of weight 2 for $\Gamma_0(Dp)$, a square-free level whose four cusps are all Atkin–Lehner. We prove, uniformly in D: the Siegel main term with closed-form local densities at D and at p; the signs $\Theta|W_p = \chi(p)\Theta$, $\Theta|W_D = -\chi(p)\Theta$, $\Theta|W_{Dp} = -\Theta$, hence the support of the cuspidal part on the newforms with $a_\varphi(D) = \chi_{-D}(p)$, $a_\varphi(p) = -\chi_{-D}(p)$ (no inert vanishing), on the oldforms from level p with $a_f(p) = -\chi_{-D}(p)$, and on the oldforms from level D only when $\chi_{-D}(p) = +1$; the Rankin–Selberg formula $\langle\Theta,\varphi\rangle = C_D\chi(p)\eta_p(\varphi)L(1,\varphi)L(1,\varphi\otimes\chi_{-D})$ with $C_D = w_D^2\sqrt{D}/(8\pi^2)$; and the growth $\|G_{1,p}^{(D)}\|^2 \sim \frac{6}{\pi(D+1)}(\log p)^2$, with the second term of the two wide cusps. The constants are those of the analytic class number formula with $h = 1$: $w_D^2 L(1,\chi_{-D})^2 = 4\pi^2/D$. Everything is verified numerically for $D = 3,7,11$ (43 newforms, Rankin–Selberg ratios within 0.5%, densities to 10^{-16}).

1. SETTING

Let D be one of the seven odd Heegner discriminants, $K = \mathbb{Q}(\sqrt{-D})$, $\chi = \chi_{-D} = \left(\frac{-D}{\cdot}\right)$ the Kronecker symbol, w_D the number of units ($w_3 = 6$, $w_D = 2$ otherwise), $N_D(x,y) = x^2 + xy + \frac{D+1}{4}y^2$ the norm form (Gram matrix $\left(\begin{smallmatrix} 2 & 1 \\ 1 & (D+1)/2 \end{smallmatrix}\right)$ of determinant D), and

$$r_D(n) = \#\{(x,y):\ N_D(x,y) = n\} = w_D \sum_{d|n} \chi(d) \quad (n \geq 1),$$

$$\theta_D(z) = \sum_{n\geq 0} r_D(n) e(nz) \in M_1(\Gamma_0(D),\chi),$$

the formula for r_D being the class number one property. The Eisenstein series $E(z,s) = \sum_{\Gamma_\infty\backslash\Gamma_0(D)} \chi(d_\gamma) j(\gamma,z)^{-1} \operatorname{Im}(\gamma z)^s$ has, at $s = 0$, the expansion $1 + \frac{2}{L(0,\chi)}\sum \sigma_\chi(n)e(nz)$ with $L(0,\chi) = 2h/w_D = 2/w_D$ (Hecke), so $E(z,0) = \theta_D(z)$ without any dimension argument. For a prime $p \neq D$ put

$$\Theta_{1,p} = \theta_D(z)\theta_D(pz) = E + G \in M_2(\Gamma_0(Dp)),$$

$$T(k) = \sum_{\alpha+p\beta=k} r_D(\alpha) r_D(\beta), \qquad a_G(k) = T(k) - M(k),$$

E the Eisenstein and $G = G_{1,p}^{(D)}$ the cuspidal part. The level Dp is square-free; $\Gamma_0(Dp)$ has the four cusps $W_Q\infty$, $Q \in \{1, D, p, Dp\}$, of widths $Dp, p, D, 1$, and no cusp of the type "denominator 2δ" of Part II. (The even discriminants $D = 4, 8$, with $\theta_4 = \theta^2$ and $\theta_8 = \theta(z)\theta(2z)$, have levels $4p$, $8p$ with non-Atkin–Lehner cusps; $D = 4$ is Parts I–III, and $D = 8$ is not treated here.) By $|_k$ we denote the weight-k slash operator normalised by $(\det\gamma)^{k/2}$, so that scalars act trivially and $f(mz) = m^{-k/2}(f|_k B_m)(z)$, $B_m = \left(\begin{smallmatrix} m & 0 \\ 0 & 1 \end{smallmatrix}\right)$.

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2. The Siegel main term

Theorem 2.1. *$M(k) = \dfrac{4\pi^2 k}{Dp}\prod_\ell \beta_\ell(k)$, where $\beta_\ell(k) = (1-\ell^{-2})\sigma_{-1}(\ell^{v_\ell(k)})$ for every prime $\ell \nmid Dp$, including $\ell = 2$, and, with $v = v_\ell(k)$,*

$$\beta_p(k) = (1+\chi(p)) - \chi(p)\Big(1+\frac{1}{p}\Big)p^{-v}, \qquad \beta_D(k) = (1+\varepsilon) - \varepsilon\Big(1+\frac{1}{D}\Big)D^{-v},$$

where $\varepsilon = \left(\frac{-p}{D}\right) = -\chi_{-D}(p)$.

Proof. The quaternary form $Q = N_D(x,y)+pN_D(z,w)$ has Gram determinant D^2p^2, and Siegel's theorem gives $r(\operatorname{gen} Q, k) = 4\pi^2 k\prod_\ell \beta_\ell(k)/\sqrt{D^2p^2}$ with $\beta_\ell(k) = \lim_a \ell^{-3a}\#\{x \bmod \ell^a : Q(x) \equiv k\}$; the genus theta series is the Eisenstein part of $\Theta_{1,p}$. For $\ell \nmid Dp$ the form is unimodular over $\mathbb{Z}_\ell$ with square discriminant, which gives the standard density; this includes $\ell = 2$ because N_D is even unimodular over $\mathbb{Z}_2$, $pN_D \cong N_D$ there (p a unit, the Arf invariant being insensitive to unit scaling), and $N_D \oplus N_D$ has Arf invariant 0, i.e. is the sum of two hyperbolic planes, whatever the class of N_D modulo 2 ($D \equiv 3$ or $7 \pmod 8$).

At $\ell = p$: the solutions with $(x,y) \not\equiv (0,0) \pmod p$ are non-singular in (x,y) (the gradient of N_D does not vanish there, N_D being non-degenerate modulo p), so each solution modulo p lifts to p^{a-1} solutions modulo p^a; modulo p the equation $N_D(x,y) \equiv t$ has $p-\chi(p)$ solutions for $p \nmid t$ (the number of elements of norm t in $\mathbb{F}_{p^2}$ or $\mathbb{F}_p\times\mathbb{F}_p$) and $1+(p-1)(1+\chi(p))$ for $p \mid t$. With (z,w) free, this gives $\beta_p(k) = 1-\chi(p)/p$ for $p \nmid k$ and the contribution $(1+\chi(p))(1-1/p)$ for $p \mid k$. The solutions with $x \equiv y \equiv 0 \pmod p$, $x = px'$, $y = py'$, satisfy $pN_D(x',y') + N_D(z,w) \equiv k/p \pmod{p^{a-1}}$, the same form with the blocks interchanged; counting (x',y') modulo p^{a-1} and the p^2 lifts of (z,w) gives the contribution $\beta_p(k/p)/p$. Hence $\beta_p(k) = (1+\chi(p))(1-1/p)+\beta_p(k/p)/p$ for $p \mid k$, which iterates to the closed form.

At $\ell = D$: since 2 is invertible, $N_D(x,y) = (x+y/2)^2 + \frac{D}{4}y^2 \cong u^2 + Dv^2$ over $\mathbb{Z}_D$, so $Q \cong u^2 + ps^2 + D(v^2+pt^2)$. The solutions with $(u,s) \not\equiv 0 \pmod D$ are non-singular in (u,s); modulo D the binary form u^2+ps^2, of discriminant $-4p$, represents $t \not\equiv 0$ exactly $D-\varepsilon$ times and $t \equiv 0$ exactly $1+(D-1)(1+\varepsilon)$ times, $\varepsilon = (-p/D)$. The solutions with $u \equiv s \equiv 0$ reduce to $D(u'^2+ps'^2)+(v^2+pt^2) \equiv k/D \pmod{D^{a-1}}$, the same form, with the factor $D^2/D^3 = 1/D$. Hence $\beta_D(k) = 1-\varepsilon/D$ for $D \nmid k$ and $\beta_D(k) = (1+\varepsilon)(1-1/D)+\beta_D(k/D)/D$ for $D \mid k$. Finally $(-p/D) = (-1/D)(p/D) = -(p/D) = -(-D/p) = -\chi_{-D}(p)$ by quadratic reciprocity, $D \equiv 3 \pmod 4$. □

For $D = 3$ this is Proposition 2.1 of the working notes; $1+1/D = 4/3$ there. The script `heegner_theta_unified.py` checks the closed forms against direct counts modulo D^a and p^a ($D = 3,7,11$, $p \le 13$; relative error $2\cdot 10^{-16}$), and the whole main term against the known cusp forms: at the level 15, $T-M = \frac{9}{2}f_{15}$; at the level 14, $T-M = \frac{2}{3}f_{14}$ ($f_{14} = \eta(z)\eta(2z)\eta(7z)\eta(14z)$); at the level 22, $T-M = 0$ identically (Section 3 explains why); at the levels $21, 33, 35, 39, 55, 77, 91, 143$ the mean square of $T-M$ grows like Y^2, as for a cusp form.

3. Atkin–Lehner signs and the spectral support

Theorem 3.1. *$\Theta_{1,p}|_2 W_p = \chi(p)\Theta_{1,p}$, $\Theta_{1,p}|_2 W_D = -\chi(p)\Theta_{1,p}$ and $\Theta_{1,p}|_2 W_{Dp} = -\Theta_{1,p}$, for any choice of the Atkin–Lehner matrices of level Dp.*

Proof. Write $W_p = \left(\begin{smallmatrix} pa & b \\ Dpc & pd\end{smallmatrix}\right)$, $pad - Dbc = 1$. Then $W_p = \gamma B_p$ with $\gamma = \left(\begin{smallmatrix} a & b \\ Dc & pd\end{smallmatrix}\right) \in \Gamma_0(D)$, so $\theta_D|_1 W_p = \chi(pd)\,\theta_D|_1 B_p = \chi(pd)p^{1/2}\theta_D(pz)$; and $B_pW_p = p\left(\begin{smallmatrix} pa & b \\ Dc & d\end{smallmatrix}\right)$ with the second factor in $\Gamma_0(D)$, so $\theta_D(p\,\cdot)|_1 W_p = p^{-1/2}\chi(d)\theta_D$. The product is $\chi(p)\chi(d)^2\Theta_{1,p} = \chi(p)\Theta_{1,p}$.

For W_D we use the transformation of θ_D under its own Atkin–Lehner matrices. The dual of the lattice $(\mathcal{O}_K, N_D)$ is $\mathcal{O}_K/\sqrt{-D}$, so Poisson summation gives $\theta_D(-1/z) = D^{-1/2}(-iz)\theta_D(z/D)$ and $\theta_D(-1/(Dz)) = -i\sqrt{D}z\,\theta_D(z)$, i.e. $\theta_D|_1 W_0 = -i\theta_D$ for $W_0 = \left(\begin{smallmatrix} 0 & -1 \\ D & 0 \end{smallmatrix}\right)$. Every matrix $W = \left(\begin{smallmatrix} Da' & b' \\ Dc' & Dd' \end{smallmatrix}\right)$ of determinant D is $W_0\gamma'$ with $\gamma' = \left(\begin{smallmatrix} c' & d' \\ -Da' & -b' \end{smallmatrix}\right) \in \Gamma_0(D)$, whence $\theta_D|_1 W = -i\chi(-b')\theta_D$. Now $W_D = \left(\begin{smallmatrix} Da & b \\ Dpc & Dd \end{smallmatrix}\right)$, $Dad - pbc = 1$, is such a matrix with $b' = b$, and $B_p W_D = \left(\begin{smallmatrix} Da & pb \\ Dc & Dd \end{smallmatrix}\right) B_p$ exhibits $\theta_D(p\,\cdot)|_1 W_D = p^{-1/2}(\theta_D|_1 W')|_1 B_p$ with W' such a matrix with $b' = pb$. Therefore $\Theta_{1,p}|_2 W_D = (-i)^2\chi(-b)\chi(-pb)\Theta_{1,p} = -\chi(p)\Theta_{1,p}$, using $(b, D) = 1$. Finally $W_{Dp} \equiv W_D W_p$ modulo $\Gamma_0(Dp)$ gives $-\chi(p)\cdot\chi(p) = -1$. The signs are checked numerically at $D = 3, 7, 11$, $p \le 13$, to 10^{-15}. □

Theorem 3.2 (Spectral support)**.** *Let φ be a newform of weight 2 and level Dp. If $\langle \Theta_{1,p}, \varphi\rangle \neq 0$ then*

$$\eta_p(\varphi) = \chi(p), \quad \eta_D(\varphi) = -\chi(p), \qquad \text{i.e.} \qquad a_\varphi(D) = \chi_{-D}(p), \quad a_\varphi(p) = -\chi_{-D}(p),$$

and the root number of φ is $+1$. For a newform f of level p and $d \mid D$, $\langle \Theta_{1,p}, f(d\,\cdot)\rangle \neq 0$ requires $\eta_p(f) = \chi(p)$, i.e. $a_f(p) = -\chi_{-D}(p)$ (root number $+1$). For a newform f of level D and $d \mid p$, $\langle \Theta_{1,p}, f(d\,\cdot)\rangle = C_D\big(1 + \chi(p)\big)L(1, f)L(1, f\otimes\chi)/d$ vanishes when $\chi_{-D}(p) = -1$.

Proof. $\langle \Theta, \varphi\rangle = \langle \Theta | W, \varphi | W\rangle$ for $W = W_p, W_D$ and Theorem 3.1 give $\chi(p)\eta_p(\varphi) = 1$ and $-\chi(p)\eta_D(\varphi) = 1$; $a_\varphi(\ell) = -\eta_\ell(\varphi)$ for $\ell \parallel Dp$, and the root number is $-\eta_D\eta_p$. For f of level p, Lemma 6.6 of Part II (with $Q = p$, $Q_1 = p$, $Q_2 = Q_3 = 1$) gives $f(d\,\cdot)|_2 W_p = \eta_p(f) f(d\,\cdot)$, and the same argument applies; W_D exchanges $f(z)$ and $f(Dz)$ up to a factor and gives no further condition. For f of level D the formula is the oldform computation in the proof of Part II, Theorem 6.7(ii), with D in the role of 4 and p in the role of n (the level of f is "2^{e}" with $M_1 = 1$, and $d \mid p$ is the "d_{o}"): the second bracket there is $\sum_{Q_2 \mid d}\chi(Q_2)$ and the third is $\prod_{\ell \mid p/d}(1 + \chi(\ell))$, so the product is $1 + \chi(p)$ in both cases $d = 1$, $d = p$. □

In contrast with $D = 4$, where every newform of level $4n$ has $\eta_4 = -1$ (Part II, Proposition 6.17), so that the newform support is empty when $\chi_{-4}(n) = -1$, here both eigenvalues of W_D occur among the newforms of level Dp, since the local component at D may be the Steinberg representation or its unramified quadratic twist: there is no inert vanishing, and for every p the newform support is the subspace $\{a_\varphi(D) = \chi(p),\ a_\varphi(p) = -\chi(p)\}$, cut down further by $L(1, \varphi)L(1, \varphi\otimes\chi) \neq 0$. This subspace is not exactly half of the new subspace: its dimension is $\frac{1}{4}(\dim S^{\mathrm{new}} + \chi(p)\operatorname{tr} W_p - \chi(p)\operatorname{tr} W_D - \operatorname{tr} W_{Dp})$, the traces taken on $S_2^{\mathrm{new}}(\Gamma_0(Dp))$, and the traces of the Atkin–Lehner involutions are sums of class numbers of size $O(\sqrt{Dp}\log Dp)$ (Yamauchi's trace formula), so it is asymptotically one quarter of the new subspace, i.e. half of the forms with root number $+1$. [Citation to be checked.] The data: for $D = 3$, $p \in \{5, 7, 11, 13, 17, 19, 23, 29, 31, 37\}$ the numbers of newform components of $G_{1,p}$ are $1, 1, 1, 2, 2, 2, 2, 3, 3, 4$ against $\dim S^{\mathrm{new}} = 1, 1, 1, 1, 3, 3, 3, 5, 5, 7$; for $D = 7$, $p \in \{5, 11, 13, 17, 19\}$: $2, 3, 3, 5, 4$; for $D = 11$, $p \in \{5, 7, 13, 17, 19\}$: $2, 3, 6, 6, 9$. At the level 77 the same three newforms carry both $G_{1,11}^{(7)}$ and $G_{1,7}^{(11)}$ ($a(7) = +1 = \chi_{-7}(11) = -\chi_{-11}(7)$, $a(11) = -1$), with the shares $0.001, 0.019, 0.981$ and $0.502, 0.495, 0.005$ respectively, in accordance with the two different twists in Theorem 4.1. The oldforms from level D exist for $D \in \{11, 19, 43, 67, 163\}$ (genus $1, 1, 3, 5, 13$) and not for $D = 3, 7$; at the level 22 ($\chi_{-11}(2) = -1$) the cuspidal part vanishes identically, $\theta_{11}(z)\theta_{11}(2z) = E$, while at the level 55 ($\chi_{-11}(5) = +1$) the component of f_{11} is $f_{11}(z) + 5f_{11}(5z)$ up to a scalar and carries 25% of the norm.

4. The Rankin–Selberg formula

Theorem 4.1. *For a newform φ of weight 2 and level Dp,*

$$\langle \Theta_{1,p}, \varphi \rangle = C_D\, \chi(p)\, \eta_p(\varphi)\, L(1,\varphi)\, L(1,\varphi\otimes\chi_{-D}), \qquad C_D = \frac{w_D}{4\pi L(1,\chi_{-D})} = \frac{w_D^2\sqrt{D}}{8\pi^2}.$$

Thus $C_3 = \frac{9\sqrt{3}}{2\pi^2} = 0.7897$, $C_D = \frac{\sqrt{D}}{2\pi^2}$ for $D \geq 7$ ($C_7 = 0.1340$, $C_{11} = 0.1680$), against $C_4 = \frac{4}{\pi^2}$.

Proof. The proof of Part II, Theorem 6.7(i) carries over: $\theta_D = E(z,0)$; the cusps of $\Gamma_0(Dp)$ over ∞ of $\Gamma_0(D)$ are $W_Q\infty$, $Q \mid p$; unfolding gives

$$I(s) = \sum_{Q\mid p} \chi(Q)\, \Gamma(s+1)(4\pi)^{-s-1} \sum_{j\geq 1} \frac{r_D(j)\, c_Q((p/Q)j)}{((p/Q)j)^{s+1}},$$

with $\chi(a)\chi(d) = \chi(Q)$ from $Qad \equiv 1 \pmod D$, and $c_Q(k) = \eta_Q(\varphi) a_\varphi(k)$. The Dirichlet series is $w_D\, L(w,\varphi)L(w,\varphi\otimes\chi)/\big(L(2w-1,\chi)\prod_{\ell\mid p}(1-\chi(\ell)\ell^{1-2w})\big)$, the local factor at $\ell = D$ being $(1-a_\varphi(D)x)^{-1} = L_D(w,\varphi)L_D(w,\varphi\otimes\chi)$ because $\chi(D) = 0$ and $\sigma_\chi(D^j) = 1$. At $s = 0$ the constant is $w_D/(4\pi L(1,\chi))$, and $\sum_{Q\mid p}\chi(Q)\eta_Q(\varphi)a_\varphi(p/Q)/(p/Q) = \chi(p)\eta_p(\varphi)(1-\chi(p)/p)$ cancels the local factor. The class number formula $L(1,\chi_{-D}) = 2\pi h/(w_D\sqrt{D})$ with $h = 1$ gives the second expression. Numerically, for the 43 newform components of $G_{1,p}^{(D)}$ with $D = 3, 7, 11$ and $p \leq 19$, the ratio of the two sides lies in $[0.9952, 1.0048]$ (the twist $\varphi\otimes\chi_{-D}$ has level D^2p and root number $+1$ in every case). □

5. The norm

Theorem 5.1. *Let $D_D(s) = \sum_k r_D(k)^2 k^{-s}$. Then $D_D(s) = w_D^2\, \zeta(s)^2 L(s,\chi)^2/\big((1+D^{-s})\zeta(2s)\big)$, and*

$$S_D(Y) := \sum_{k\geq 1} \frac{r_D(k)^2}{k} e^{-k/Y} = \frac{12}{D+1}(\log Y)^2 + c_1^{(D)} \log Y + c_2^{(D)} + O(Y^{-1/2}),$$

$$c_1^{(D)} = \frac{24}{D+1}\Big(\gamma + 2\frac{L'}{L}(1,\chi) - 2\frac{\zeta'}{\zeta}(2) + \frac{\log D}{D+1}\Big),$$

with $\frac{L'}{L}(1,\chi_{-D}) = \gamma + \log 2 + \log\pi - \frac{w_D}{2}\sum_{a=1}^{D-1}\chi(a)\log\Gamma(a/D)$. Consequently the regions above height 1 at the two wide cusps give

$$\|G_{1,p}^{(D)}\|^2 \;\geq\; \frac{1}{4\pi}\Big(S_D\Big(\frac{Dp}{4\pi}\Big) + S_D\Big(\frac{p}{4\pi}\Big)\Big) - O(\log p) = \frac{6}{\pi(D+1)}(\log p)^2 + b_D \log p + O(1),$$

$$b_D = \frac{6}{\pi(D+1)}\log\frac{D}{16\pi^2} + \frac{c_1^{(D)}}{2\pi} \qquad (b_3 = 0.713,\ b_7 = 0.207,\ b_{11} = 0.133),$$

and $\|G_{1,p}^{(D)}\|^2 \sim \frac{6}{\pi(D+1)}(\log p)^2$ provided the upper bound of Part II, Theorem 6.14, carries over (four Atkin–Lehner classes, no odd class).

Proof. $r_D/w_D = \sigma_\chi$ is multiplicative with $\sigma_\chi(\ell^j) = j+1$, $\frac{1}{2}(1+(-1)^j)$, 1 for $\chi(\ell) = 1, -1, 0$, so the Euler factor of $\sum\sigma_\chi(k)^2k^{-s}$ at $\ell \neq D$ is that of $\zeta^2L^2/\zeta(2s)$, and at $\ell = D$ it is $(1-x)^{-1}$ against $(1+x)/(1-x)$ for $\zeta^2L^2/\zeta(2s)$, whence the factor $(1+D^{-s})^{-1}$. At $s = 1$: $w_D^2 L(1,\chi)^2 \cdot \frac{D}{D+1}\cdot\frac{6}{\pi^2} = \frac{4\pi^2}{D}\cdot\frac{D}{D+1}\cdot\frac{6}{\pi^2} = \frac{24}{D+1}$ by the class number formula, so $D_D(1+w) = \frac{24}{D+1}w^{-2}(1+a_1w+\dots)$ with $a_1 = 2\gamma + 2\frac{L'}{L}(1,\chi) - 2\frac{\zeta'}{\zeta}(2) + \frac{\log D}{D+1}$, and the residue computation of Part II, Lemma 6.10 gives S_D. The value of $\frac{L'}{L}(1,\chi)$ follows from the functional equation $\Lambda(s) = (D/\pi)^{s/2}\Gamma(\frac{s+1}{2})L(s,\chi) = \Lambda(1-s)$ (root number $+1$ for real odd χ) and $L(s,\chi) =$

$D^{-s}\sum_a \chi(a)\zeta(s,a/D)$ with $\zeta'(0,a)=\log\Gamma(a)-\frac12\log 2\pi$: $\frac{L'}{L}(1)=-\log\frac{D}{\pi}-\frac12\psi(\frac12)-\frac12\psi(1)-\frac{L'}{L}(0)$ and $\frac{L'}{L}(0)=-\log D+\frac{w_D}{2}\sum_a\chi(a)\log\Gamma(a/D)$, since $L(0,\chi)=2/w_D$. The lower bound for the norm is Part II, Lemma 6.9 at the cusps $W_1\infty$ and $W_D\infty$ (widths Dp and p), which carry $\pm G_{1,p}$ by Theorem 3.1, with the correction by the main term and by the coefficients $k\ge p$ bounded as in Part II, Theorem 6.5(iii), and $\log\frac{Dp}{4\pi}+\log\frac{p}{4\pi}=2\log p+\log\frac{D}{16\pi^2}$. □

The growth constant is $\frac{6}{\pi(D+1)}$, decreasing in D (0.4775, 0.2387, 0.1592 for $D=3,7,11$, against $\frac1\pi=0.3183$ for $D=4$, where the factor at the ramified prime is $\frac23$ instead of $\frac{D}{D+1}$); the value $\frac{\sqrt{D}}{2\pi}$ proposed for it is incorrect. What the class number formula fixes is the product $w_D^2L(1,\chi_{-D})^2=4\pi^2/D$, which makes the residue of D_D independent of w_D; the remaining factor $D/(D+1)$ is the Euler factor at the ramified prime, where $\sigma_\chi(D^j)=1$.

6. Numerical results

D	w_D	$L(1,\chi)$	$\frac{L'}{L}(1,\chi)$	C_D	$\frac{12}{D+1}$	$c_1^{(D)}$	$c_2^{(D)}$	$\frac{6}{\pi(D+1)}$	fit a
3	6	0.604600	0.368282	0.789720	3	16.370	13.059	0.4775	0.4815
7	2	1.187410	0.015636	0.134035	1.5	5.975	1.450	0.2387	0.2408
11	2	0.947226	−0.084218	0.168022	1	3.497	0.492	0.1592	0.1601

($L(1,\chi)$ and $\frac{L'}{L}(1,\chi)$ agree with mpmath to all digits shown; $c_2^{(D)}$ from $S_D(Y)$ at $Y=10^4,10^5$; "fit a" is the coefficient of $(\log p)^2$ in a three-parameter fit of $\Lambda_1(G_{1,p})$ over $p\in\{101,1009,10007,100003\}$.) The sandwich $\Lambda_1\le\|G\|^2\le\Lambda_{\sqrt3/2}$ over the four cusps, and the Rankin–Selberg estimate where it has converged ($p\le1009$):

	$D=3$			$D=7$			$D=11$		
p	RS	Λ_1	$\Lambda_{\sqrt3/2}$	RS	Λ_1	$\Lambda_{\sqrt3/2}$	RS	Λ_1	$\Lambda_{\sqrt3/2}$
101	11.893	11.621	12.513	5.437	5.294	5.774	4.092	3.969	4.363
1009	26.452	25.966	27.300	12.556	12.263	13.013	9.040	8.783	9.402
10007		45.385	47.151		21.768	22.785		15.277	16.117
100003		69.921	72.120		33.833	35.116		23.479	24.540

The exact norm by the scalloped Parseval formula of the working notes applies verbatim (four Atkin–Lehner classes) and has not been run for $D\ne4$.

7. The remaining odd discriminants, and the even ones

7.1. $D=19$ **and** $D=43$ (`results/heegner_theta_unified_19_43_partial.txt`)**.** The script was run for $D=19,43,67,163$; the system stopped it for lack of memory during the norm scan of $D=43$, so that $D=19$ is complete, $D=43$ is complete up to the spectral support, and $D=67,163$ were not reached (they are run one at a time by `python heegner_theta_unified.py 67`, etc.; for these two the Hecke decomposition is beyond the Prony method, $\dim S_2(\Gamma_0(Dp))\ge60$, and only the densities, the Atkin–Lehner signs and the norm scan would be checked). For $D=19$ ($N_{19}=x^2+xy+5y^2$, $L(1,\chi)=0.720731$, $\frac{L'}{L}(1,\chi)=-0.084914$, $C_{19}=0.220824$, $\frac{12}{D+1}=0.6$, $c_1^{(19)}=2.033$, $\frac{6}{\pi(D+1)}=0.0955$): the densities agree with the counts to $2\cdot10^{-16}$, the three Atkin–Lehner signs hold for $p\le13$, every newform component of $G_{1,p}$ for $p\in\{5,7,11,13,17\}$ (23 newforms) has $a(19)=\chi_{-19}(p)$, $a(p)=-\chi_{-19}(p)$, the Rankin–Selberg ratios lie in $[0.9965,1.0047]$, and the norm scan gives

p	RS	Λ_1	$\Lambda_{\sqrt{3}/2}$	$\frac{6}{20\pi}(\log p)^2$	$\Lambda_1/(\log p)^2$
101	3.402	3.293	3.637	2.034	0.155
1009	6.533	6.502	7.023	4.569	0.136
10007		10.733	11.429	8.102	0.127
100003		15.975	16.843	12.657	0.121

with the fit $\Lambda_1 = 0.0961(\log p)^2 + 0.290\log p - 0.09$ over $p \geq 100$, against 0.0955. For $D = 43$ ($N_{43} = x^2 + xy + 11y^2$, $L(1,\chi) = 0.479088$, $\frac{L'}{L}(1,\chi) = 0.249482$, $C_{43} = 0.332204$, $\frac{12}{D+1} = 0.2727$, $c_1^{(43)} = 1.255$, $\frac{6}{\pi(D+1)} = 0.0434$): densities and Atkin–Lehner signs as above; for $p = 5, 7, 11$ (levels $215, 301, 473$, $\dim S_2 = 21, 27, 43$) the decomposition resolves 13 newform components, all obeying the law, with Rankin–Selberg ratios in $[0.9936, 1.0038]$ (at $p = 5$ one numerically contaminated duplicate of a component, with $a(43) = -1.0001$, is mislabelled by the script; its clean copy obeys the law); for $p = 13, 17, 19$ ($\dim S_2 \geq 49$) the Prony recurrence of order ≤ 11 does not separate the span, and nothing is tested. The predicted growth constants for the two largest discriminants are $\frac{6}{68\pi} = 0.0281$ and $\frac{6}{164\pi} = 0.01165$, the latter 41 times smaller than for $D = 3$; Section 8 says what this suppression does and does not mean.

7.2. **The even discriminants $D = 4, 8$.** For $D = 4$ ($\theta_4 = \theta^2$, $w_4 = 4$) and $D = 8$ ($\theta_8(z) = \sum e((x^2 + 2y^2)z)$, $w_8 = 2$) the theta series are again the Eisenstein series $E_1(z, \chi_{-D})$ of weight one, since $h = 1$, and the level Dp has, besides the Atkin–Lehner cusps $W_Q\infty$, $Q \mid Dp$, the cusps of denominator 2δ ($D = 4$) and $2\delta, 4\delta$ ($D = 8$) of Part II, Section 6.6. The two ingredients that fix the constants are the same: $r_D = w_D\sigma_\chi$ and the class number formula $w_D^2 L(1,\chi_{-D})^2 = 4\pi^2/D$. Hence the Rankin–Selberg constant is $C_D = w_D/(4\pi L(1,\chi_{-D})) = w_D^2\sqrt{D}/(8\pi^2)$ for all nine discriminants ($C_4 = 4/\pi^2$, Part II; $C_8 = \sqrt{2}/\pi^2$, with the same proof once the cusps of $\Gamma_0(8p)$ over the cusp ∞ of $\Gamma_0(8)$ are listed, which we have not written out), and the residue of $D_D(s) = \sum r_D(k)^2 k^{-s}$ at $s = 1$ is $w_D^2 L(1,\chi)^2 \cdot \frac{6}{\pi^2} \cdot \prod_{\ell\mid D}(1 + \ell^{-1})^{-1} = 24/\psi(D)$, where the Euler factor at each ramified prime is $(1 + \ell^{-s})^{-1}$ because $\sigma_\chi(\ell^j) = 1$ there, and $\psi(D) = D\prod_{\ell\mid D}(1 + 1/\ell)$ is the Dedekind function. The two wide Atkin–Lehner cusps (widths Dp and p, which carry $\pm G_{1,p}$ by the same computation as in Theorem 3.1, using $\theta_D(-1/(Dz)) = -i\sqrt{D}z\,\theta_D(z)$ for $D = 8$ as well) therefore give, uniformly for the nine discriminants,

$$\|G_{1,p}^{(D)}\|^2 \;\geq\; \frac{6}{\pi\,\psi(D)}(\log p)^2 + O(\log p) = \frac{2}{\mathrm{vol}(\Gamma_0(D)\backslash\mathbb{H})}(\log p)^2 + O(\log p),$$

since $\mathrm{vol}(\Gamma_0(D)\backslash\mathbb{H}) = \frac{\pi}{3}\psi(D)$; this is $\frac{1}{\pi}$ for $D = 4$ (Part II, Theorem 6.14, where it is the asymptotic), $\frac{1}{2\pi}$ for $D = 8$, and $\frac{6}{\pi(D+1)}$ for the odd D. The growth constant of the cusp form is thus the reciprocal of the volume of the modular curve of the *discriminant*, not of the level: the natural normalisation of $\|G\|^2$ against $\mathrm{vol}(\Gamma_0(Dp)\backslash\mathbb{H})$, which the Rankin–Selberg mean square $\sum_{k\leq Y}|a(k)|^2/k \sim 16\pi^2\|G\|^2 Y/\mathrm{vol}(\Gamma_0(Dp)\backslash\mathbb{H})$ uses, makes the mean square of $r_D(\alpha)r_D(\beta)$ over $\alpha + p\beta \leq Y$ equal to $2(\log Y)^2 \cdot \frac{Y}{p} \cdot \frac{\mathrm{vol}(\Gamma_0(D)\backslash\mathbb{H})}{\mathrm{vol}(\Gamma_0(Dp)\backslash\mathbb{H})}\cdots$, and the index $[\Gamma_0(D) : \Gamma_0(Dp)] = p + 1$ cancels against the $1/p$ of the sparse sequence $p\beta$.

8. Does a large discriminant help the sieve?

For the sieve problem of Parts I–III with r_D in place of r, i.e. the sequence $r_D(n)r_D(N-n)$, the norm $\|G_{m,m'}^{(D)}\|^2 \sim \frac{6}{\pi\psi(D)}(\log mm')^2$ enters the remainder through the uniform bound $|a_G(N)| \ll N^{1/2+\varepsilon}\|G\|$ (Part I, Theorem 2.3, whose proof is independent of D), and it decreases like $1/D$. This does not help. First, the main term decreases as well: the density of the sequence is governed by r_D, whose mean value is $w_D L(1,\chi_{-D}) = 2\pi/\sqrt{D}$, so the main term $X_N = \sum_n r_D(n)r_D(N -$

$n) \asymp \frac{4\pi^2}{D} N\,\mathfrak{S}_D(N)$ is of size N/D, while the remainder for one pair (d,e) of sieve moduli is $\ll N^{1/2+\varepsilon}(\log de)/\sqrt{\psi(D)}$; the ratio remainder/main is therefore $\asymp \sqrt{D}\cdot N^{-1/2}(\log de)$ for prime D, *larger* by $\sqrt{D}$, not smaller. Second, and decisively, the level of distribution is an exponent, and D is fixed: the proof of the level $1/2$ (Part I, Theorem 3.2) sums $|a_{G_{d,e}}(N)|$ over $de \le N^{1/2-\varepsilon}$ and loses only constants and N^{ε} in D, so the level is $1/2$ for every D, no more and no less; and the sieve dimension is 2 for every D, since $\sum_{p\le z}\omega_D(p)/p = 2\log\log z + O(1)$ with $\omega_D(p) = 2(1+\chi_{-D}(p))$ on average. What a large discriminant changes is only the constant in the final count ($\asymp N/(D(\log N)^2)$ in place of $N/(\log N)^2$) and, for the odd D, the shape of the fundamental domain (no cusps of denominator 2δ), which simplifies the analysis of the norm but has no bearing on the dispersion problem of Section 9.

9. The Eisenstein dispersion form

Here $D = 3$, m, m' coprime square-free integers prime to $3N$, and $T(m,m';N) = \sum_{m\alpha+m'\beta=N} r_3(\alpha)r_3(\beta)$. We follow Part III, Section 3 (in its corrected form: the Kloosterman sums there are $S(\overline{16m}N, |h|^2; m'k)$, a Kloosterman sum depending on its arguments only through their product, so that the inverse of $16m$ modulo the modulus cannot be removed).

Lemma 9.1 (The circle modulo c)**.** *Let $(c,3)=1$, c odd or even, $(a,c)=1$, $h\in\mathbb{Z}^2$, and $|h|_3^2 = h_1^2 - h_1h_2 + h_2^2$ the norm of the dual (hexagonal) lattice. With c'^{*} the product of the primes dividing c' to an odd power and $\chi = \chi_{-3}$ (so $\chi(2) = -1$),*

$$F_c(a;h) := \sum_{\substack{x,y \bmod c \\ N_3(x,y)\equiv a \ (c)}} e\Big(\frac{h_1x+h_2y}{c}\Big) = \sum_{g\mid(c,h_1,h_2)} g\,\chi\Big(\Big(\frac{c}{g}\Big)^{*}\Big) S\Big(a,\ \overline{3}\,\Big|\frac{h}{g}\Big|_3^2;\ \frac{c}{g}\Big).$$

Proof. As in Part III, Lemma 3.1. Detect the congruence by $\frac{1}{c}\sum_{t \bmod c} e(t(N_3(x,y)-a)/c)$; for $(t,c) = g$ the sum over (x,y) vanishes unless $g \mid h$, and for $(t,c)=1$ one completes the square: with the Gram matrix $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $N_3 = \frac{1}{2}x^{\mathsf T}Ax$, the substitution $x \mapsto x - \bar t A^{-1}h$ (legitimate modulo c since $\det A = 3$ is invertible, also at 2) gives $\sum_{x,y} e((tN_3(x,y)+h\cdot x)/c) = e(-\bar t\, h^{\mathsf T}A^{-1}h/(2c))\,G_A(t;c)$ with $h^{\mathsf T}A^{-1}h/2 = |h|_3^2/3$, and $G_A(t;c) = \sum_{x,y} e(tN_3(x,y)/c) = G(t;c)\,G(3\bar 4 t;c)$ for odd c after diagonalising $N_3 = (x+y/2)^2 + \frac{3}{4}y^2$, which equals $c\,\chi((c)^{*})$ by the evaluation of the quadratic Gauss sums prime by prime, the twisting Jacobi symbols of the Chinese remainder theorem squaring to 1; at the prime 2 the form is unimodular and $G_A(t;2^\nu) = (-1)^\nu 2^\nu = \chi((2^\nu)^{*})2^\nu$. The script `eisenstein_circle_check.py` verifies the identity for all a, h and all $c \le 40$ coprime to 3, odd and even (285 902 cases). □

The decomposition. Since the ideal $(\sqrt{-3})$ is principal of norm 3, $r_3(3\beta') = r_3(\beta')$, and the terms of $T(m,m';N)$ with $3 \mid \beta$ are $T(m,3m';N)$: with $T^{\flat}(m,m';N) = \sum_{3\nmid\beta} r_3(\alpha)r_3(\beta)$,

$$a_{G_{m,m'}}(N) - a_{G_{m,3m'}}(N) = T^{\flat}(m,m';N) - M^{\flat}(m,m';N),$$

the analogue of $a_G(N) - a_G(N/2) = T^{\mathrm{odd}} - M^{\mathrm{odd}}$ of Part III (there $r(2n) = r(n)$ for the ideal $(1+i)$). In $T^{\flat}$, $r_3(\beta)\neq 0$ forces $\beta \equiv 1 \pmod 3$ (every prime $\equiv 2 \pmod 3$ divides β to an even power), so $\chi(\beta/k) = \chi(k)$ and $r_3(\beta) = 12\sum_{k\mid\beta,\,k<\sqrt{\beta}}\chi(k) + 6\chi(\sqrt{\beta})[\beta = \square]$; the divisors k are

prime to 3 and to m (a common prime would divide N) but may be even. Hence

$$T^\flat(m,m';N)=12\sum_{\substack{k<\sqrt{N/m'}\\ 3\nmid k}}\chi(k)\,\mathcal{N}_k+6\sum_j\chi(j)\,r_3^\flat\Big(\frac{N-m'j^2}{m}\Big),$$

$$\mathcal{N}_k=\#\Big\{(x,y):\ N_3(x,y)\equiv N\bar m\ (m'k),\ mN_3(x,y)\not\equiv N\ (3),\ 0<N_3(x,y)<Y_k\Big\},$$

$Y_k=(N-m'k^2)/m$. The condition modulo 3 is the analogue of the parity condition $x+y$ odd of Part III: Poisson summation on the classes modulo $3c$, $c=m'k$, gives

$$\mathcal{N}_k=\frac{Y_k}{(3c)^2}\sum_{h\in\mathbb{Z}^2}\hat w\Big(\frac{h\sqrt{Y_k}}{3c}\Big)\,\epsilon_3(h)\,F_c(N\bar m;\ \bar 3h),$$

where $\bar 3h$ is the Chinese-remainder twist and $\epsilon_3(h)$ is the Fourier transform of the admissible classes modulo 3: since $N_3(x,y)\equiv(x-y)^2\pmod 3$, the condition $mN_3\not\equiv N\pmod 3$ is $x\equiv y$ if $N\bar m\equiv 1$, $x\not\equiv y$ if $N\bar m\equiv 0$, and empty if $N\bar m\equiv 2\pmod 3$, so that

$$\epsilon_3(h)=3\,[h_1+h_2\equiv 0\ (3)],\qquad 9[h\equiv 0\ (3)]-3[h_1+h_2\equiv 0\ (3)],\qquad 9,$$

respectively. As $|h|_3^2\equiv(h_1+h_2)^2\pmod 3$, the condition $h_1+h_2\equiv 0\pmod 3$ is $3\mid|h|_3^2$: exactly the condition $h_1\equiv h_2\pmod 2$, i.e. $2\mid|h|^2$, of Part III, with the ramified prime of the field in the role of 2. By Lemma 9.1 the leading term $g=1$ of $F_c(N\bar m;\bar 3h)$ is $\chi((m'k)^*)S(N\bar m,\overline{27}|h|_3^2;m'k)$, and $\chi(k)\chi((m'k)^*)=\chi(m')\chi(k/k^*)=\chi(m')$ for square-free m', a constant sign for each m' (in Part III, $m'\equiv 1\pmod 4$ made it $+1$). We have:

Proposition 9.2. *With the notation above, the leading term ($g=1$) of $a_{G_{m,m'}}(N)-a_{G_{m,3m'}}(N)$ is*

$$12\,\chi(m')\sum_{\substack{k<\sqrt{N/m'}\\ 3\nmid k}}\frac{Y_k}{(3m'k)^2}\sum_{h\neq 0}\hat w\Big(\frac{h\sqrt{Y_k}}{3m'k}\Big)\,\epsilon_3(h)\,S\big(\overline{27m}\,N,\ |h|_3^2;\ m'k\big),$$

and the remainder sum of the sieve at the level N^θ has as its leading term the trilinear form $\sum_{m'}\Lambda'_{m'}\sum_{c\equiv 0\,(m')}\sum_m\Lambda_m\sum_j b_{m,j}(c)S(\overline{27m}N,j;c)$ over the moduli $c=m'k$ prime to 3, odd or even, with $j=|h|_3^2\ll mm'$ a norm from $\mathbb{Z}[\zeta_3]$; by the Chinese remainder theorem its m-th piece is the bilinear form in the two moduli

$$\mathfrak{B}_m^{(3)}=\sum_{m'}\Lambda'_{m'}\sum_{k\le K}\sum_h w_{k,m'}(h)\,S\big(\overline{27m}N\bar k,\ |h|_3^2\bar k;\ m'\big)\,S\big(\overline{27m}N\overline{m'},\ |h|_3^2\overline{m'};\ k\big),$$

the inverse of $27m$ taken modulo m' and modulo k respectively.

Comparison with the Gaussian case. Nothing in the structure changes. (i) The fixed argument is again the single integer N, carried by a unit ($\overline{27m}$ in place of $\overline{16m}$) modulo the modulus, so the obstacle (a) of Part III, Section 3.4, the absence of an average over it, is the same, and so is the obstruction to the Kuznetsov formula. (ii) The sizes are the same: $|h|\ll m'k\sqrt{m/N}$, so the h-box has side $H\le\sqrt{mm'}$, below $\sqrt{q}$ for every modulus $q=[c_1,c_2]$ of a Poisson step, and the Cauchy–Schwarz arrangements (A1), (A2) of Part III, Section 4 give the same diagonals ($E^{1/4}N^{3/4}$ and $\sqrt{EN}$); there is no tighter grouping, because the groupings depend only on the lengths of the variables and on the Weil mean square of Kloosterman sums over the dual lattice, which is the same for the hexagonal lattice. (iii) The modulus $m'k$ now ranges over all k prime to 3, even k included: more moduli, no gain. (iv) The 2-adic "impurity" of Part III (the cusps of denominator 2δ, the parity condition) has moved to the prime 3: the condition $3\nmid\beta$ produces $\epsilon_3(h)$ and the dual condition $3\mid|h|_3^2$, as $2\nmid\beta$ produced $h_1\equiv h_2\pmod 2$. This is not an accident:

the ramified prime of the field is where the ideal of norm ℓ is principal, and removing it from the divisor sum of $r_D(\beta)$ is what makes the divisors $k < \sqrt{\beta}$ pair with β/k under χ. The clean level $3p$ matters for the holomorphic side (Sections 2–5), not for the dispersion form; a level beyond $1/2$ needs the bilinear saving over (m', k) of Part III, Section 4.5, in the Eisenstein setting as in the Gaussian one.

Scripts

`code/eisenstein_circle_check.py` (Lemma 9.1, and the Kloosterman product identity behind the correction of Part III; output in `results/eisenstein_circle_check.txt`).

Script

`code/heegner_theta_unified.py` (self-contained; Kronecker symbol, r_D, T, M, counts of local densities, Atkin–Lehner check at a base point, Hecke decomposition by T_2 then T_3, T_5, T_7, T_{13} where eigenvalues coincide, central values by the smoothed functional equation, norms by Rankin–Selberg and by the cusp sums); output in `results/heegner_theta_unified.txt`. Floating point.